

PP Lecture 5 PDC

Part 1: Past Papers

MIDTERM 1 / SESSIONAL 1 SPRING 2023 / FALL 2022 / SPRING 2025 • QUESTION 1

Q: Calculate (a) diameter, (b) arc connectivity, (c) cost (number of links), and (d) bisection width for the following static interconnection architectures:

- i. 4x4 two-dimensional mesh with wraparound links (2D Torus)**
- ii. 4x4 two-dimensional mesh with no wraparound links**
- iii. Four-dimensional hypercube (size = 16 Nodes)**
- iv. Complete binary tree of 3 levels (15 nodes)**

A:

- i. 4x4 2D Mesh with wraparound links (Torus, p=16):**

$$\text{Diameter} = 2 \cdot \lfloor \sqrt{p}/2 \rfloor = 2 \cdot 2 = 4$$

$$\text{Arc Connectivity} = 4$$

$$\text{Cost} = 2p = 2 \times 16 = 32 \text{ links}$$

$$\text{Bisection Width} = 2\sqrt{p} = 2 \times 4 = 8$$

- ii. 4x4 2D Mesh with no wraparound links (p=16):**

$$\text{Diameter} = 2(\sqrt{p} - 1) = 2(4 - 1) = 6$$

$$\text{Arc Connectivity} = 2$$

$$\text{Cost} = 2(p - \sqrt{p}) = 2(16 - 4) = 24 \text{ links}$$

$$\text{Bisection Width} = \sqrt{p} = \sqrt{16} = 4$$

- iii. 4D Hypercube (p=16 nodes):**

$$\text{Diameter} = \log_2(p) = \log_2(16) = 4$$

$$\text{Arc Connectivity} = \log_2(p) = 4$$

$$\text{Cost} = (p \log_2 p) / 2 = (16 \times 4) / 2 = 32 \text{ links}$$

$$\text{Bisection Width} = p / 2 = 16 / 2 = 8$$

- iv. Complete Binary Tree of 3 levels (p=15 nodes):**

$$\text{Diameter} = 2 \log_2((p + 1)/2) = 2 \log_2(8) = 6$$

$$\text{Arc Connectivity} = 1$$

$$\text{Cost} = p - 1 = 15 - 1 = 14 \text{ links}$$

$$\text{Bisection Width} = 1$$

MIDTERM 1 / SESSIONAL 1 SPRING 2025 • QUESTION 2

Q: Derive generalized expressions for arc-connectivity, diameter, bisection-width, and link cost of a 2D mesh (without wraparound) having M rows and N columns.

A:

Arc Connectivity = 2 (each internal arc connects two nodes)

Diameter = $(M - 1) + (N - 1) = M + N - 2$

Bisection Width = $\min(M, N)$

Cost (No. of links) = $(N - 1)M + (M - 1)N = 2MN - (M + N)$

MIDTERM 1 / SESSIONAL 1 SPRING 2025 / SPRING 2023 • QUESTION 3

Q: For a 4x4 / 8x8 Multistage Omega network connecting p processors to p memory modules:

(a) Draw the 4x4 Omega network diagram for processors P_0 – P_3 and memory modules M_0 – M_3 .

(b) Explain how P_3 will access M_2 in terms of routing.

(c) How many switching stages are required for this network?

(d) Can P_1 to M_2 and P_3 to M_3 communications happen concurrently? Why or why not?

(e) How many switching nodes will be required for this network?

A:

(a) The network consists of 2x2 switching nodes arranged in stages connected via perfect shuffle interconnections.

(b) Routing P_3 to M_2 : Destination M_2 in binary is 10. At Stage 1 (using the 1st bit 1), the switch routes the message to its 1st (lower/cross) output port. At Stage 2 (using the 2nd bit 0), the switch routes to its 0th (upper/pass-through) output port to reach M_2 .

(c) Switching Stages = $\log_2(p) = \log_2(4) = 2$ stages (for 8x8, $\log_2(8) = 3$ stages).

(d) Concurrent Communication ($P_1 \rightarrow M_2$ and $P_3 \rightarrow M_3$): Not possible. A conflicting link exists on the path between the switches, as Omega networks are blocking networks.

(e) Switching Nodes = $(p / 2) \times \log_2(p) = (4 / 2) \times \log_2(4) = 4$ switching nodes (for 8x8, $(8/2) \times 3 = 12$ nodes).

QUIZ 2 SPRING 2024 • QUESTION 4

Q:

- (1) What is the name of the topology shown in the diagram (a 6-node completely connected graph)?
- (2) Define bisectional bandwidth of a topology.
- (3) What is the bisectional bandwidth of the above topology having n nodes?
- (4) Prove the expression found in Question 3.

A:

- (1) Completely-Connected Network (or Complete Graph / Fully Connected Topology).
- (2) Bisectional Bandwidth is the minimum volume of communication allowed across any cut that partitions the network into two equal halves (bisection width $\times$ channel bandwidth).
- (3) Bisectional Bandwidth = $n^2 / 4$ (assuming unit channel bandwidth).
- (4) Proof: Partition n nodes into two equal sets of $n/2$ nodes each. Since every node in one set has a direct link to every node in the other set, the total number of links crossing the bisection is $(n/2) \times (n/2) = n^2 / 4$.

FINAL EXAM SPRING 2026 • QUESTION 5

Q:

- (a) State the following properties of an 8×32 2D mesh with 256 nodes: (i) diameter, (ii) arc-connectivity, and (iii) bisection width. Then identify which two of the three properties stated above most directly constrain all-to-all broadcast performance and briefly state why.
- (b) A colleague suggests replacing the 8×32 mesh with a hypercube of 256 processes. Name one property of the hypercube that would reduce communication cost for this workload, and one practical reason the team might still prefer the mesh.

A:

- (a) For an 8×32 2D Mesh ($M=8$, $N=32$, $p=256$):
- (i) Diameter = $(M-1) + (N-1) = 7 + 31 = 38$
 - (ii) Arc-connectivity = 2
 - (iii) Bisection width = $\min(M, N) = 8$

Constraining Properties: Bisection width and diameter most directly constrain all-to-all broadcast performance because bisection width limits the available cross-section bandwidth for heavy data exchange, while diameter determines the maximum latency/hops for messages to reach distant nodes.

(b) Hypercube vs Mesh:

- Hypercube Advantage: Much smaller diameter ($\log_2(256) = 8$ vs 38) and significantly higher bisection width ($256/2 = 128$ vs 8), reducing overall communication hops and bottleneck contention.
- Practical Reason to Prefer Mesh: Simple physical layout/wiring in 2D space, constant node degree (4 vs 8), and much lower hardware implementation cost.

MIDTERM 1 / FALL 2022 / SESSIONAL 1 • QUESTION 6

Q: Compare Graph 1 (3D Hypercube with 8 nodes) and Graph 2 (4x4 2D Torus / Wraparound Mesh with 16 nodes) in terms of diameter and arc connectivity. What does it tell about their performance and reliability?

A:

3D Hypercube ($p=8$): Diameter = $\log_2(8) = 3$, Arc connectivity = 3.

4x4 2D Torus ($p=16$): Diameter = $2 \times \lfloor \sqrt{16}/2 \rfloor = 4$, Arc connectivity = 4.

Analysis: The hypercube offers lower diameter (better latency per node count), whereas the torus provides higher arc connectivity, offering more redundant routes in the event of link failures.

MIDTERM 1 / SPRING 2023 • QUESTION 7

Q: What is Fat Tree Routing? Draw a Fat tree having 4 levels with 8 leaf nodes. Provide a comparative analysis with a star topology with the same number of nodes in terms of cost, diameter, bisection width, and arc connectivity.

A:

Fat Tree Definition: A tree-structured network where link capacity/bandwidth increases towards the root (upper levels) to eliminate root bottlenecks.

Comparison:

Star: Diameter = 2, Bisection Width = 1, Arc Connectivity = 1, Cost = $p - 1$.

Fat Tree: Diameter = $2 \log_2((p+1)/2)$, Bisection Width = $p/2$ (at root), Arc Connectivity > 1 at upper levels, Cost = $n \times 2^n$.

Verdict: Star has lower cost and smaller diameter but suffers from a severe central bottleneck. Fat tree manages heavy traffic smoothly due to distributed bandwidth at higher levels.

QUIZ 5 SPRING 2025 • QUESTION 8

Q: When we use E-cube routing in a hypercube, why is the worst-case cost of circular shift $t_s + m \cdot t_w$ (without any multiplicative factor)?

A: E-cube routing in d dimensions provides 2^d un-contended, concurrent communication paths. For a d -dimensional hypercube, a transmission schedule exists such that all 2^d nodes send messages simultaneously over disjoint paths for any shift distance q , completing all transfers in a single step without channel contention.

Q:

i. In a Bus network, the number of shared connections is equal to:

A) 1 B) n C) $\log_2 n$ D) n^2

ii. Multi-stage Omega network has _____ switching nodes:

a) $\log p$ b) $(p/2) \times \log p$ c) $p \times \log p$ d) p^2

iii. A hypercube of dimension d has:

a) d nodes b) 2^d nodes c) $2d$ nodes d) None

iv. In hypercube communication loops, the expression $(my_id \text{ XOR } 2^i)$ is used to:

a) Decide the ID of the neighbor for message exchange in that iteration

b) Decide message size

c) Decide total dimensions

d) Decide farthest node ID

v. Would you agree that a multi-stage network is an example of an interconnection network where none of the data paths are shared?

vi. When calculating message passing costs, t_w is affected by:

a) Bandwidth of the links b) Switch latencies c) Number of hops d) Headers

A:

i. B) n (shared bus line accessed by n processors)

ii. b) $(p/2) \times \log p$

iii. b) 2^d nodes

iv. a) To decide the ID of the neighbor with which we will exchange messages in this iteration

v. No. Multi-stage networks share internal switching paths (blocking). Crossbar networks have non-shared data paths.

vi. a) Bandwidth of the links ($t_w = 1 / \text{bandwidth}$)

Part 2: Book Questions

Q: [Problem 2.8] Why is it difficult to construct a true shared-memory computer? What is the minimum number of switches for connecting p processors to a shared memory with b words (where each word can be accessed independently)?

A:

It is difficult because connecting p processors to b independent memory words requires a completely non-blocking crossbar switch network. As p and b grow, switch complexity grows as $\Theta(p \cdot b)$. Physical wiring constraints, pin limitations, and fan-in/fan-out delays make large crossbars unscalable.

The minimum number of switching nodes required is $\Theta(p \cdot b)$.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 2

Q: [Problem 2.10] The Butterfly network is an interconnection network composed of $\log p$ levels (as the omega network). In a Butterfly network, each switching node i at a level l is connected to the identically numbered element at level $l + 1$ and to a switching node whose number differs from itself only at the l -th most significant bit. Show the equivalence of a Butterfly network and an omega network.

A: An Omega network consists of $\log p$ stages of perfect shuffle interconnections feeding into 2×2 switches. By permuting and relabeling the positions of the switching nodes at each intermediate stage, the perfect shuffle connections at stage l map directly to the straight and crossover links at level l of the Butterfly network, proving topological equivalence.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 3

Q: [Problem 2.11] Consider an omega network connecting p processors to p memory banks. How many distinct permutation functions exist?

How many of these functions result in non-blocking communication?

What is the probability that an arbitrary function will result in non-blocking communication?

A:

There are $p!$ distinct permutation functions.

The network has $(p/2) \log p$ switches, each with 2 states. The total number of distinct state configurations is $2^{(p/2) \log p} = p^{p/2}$. Thus, at most $p^{p/2}$ permutations are non-blocking.

Probability = $p^{p/2} / p!$, which rapidly approaches 0 as p grows large.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 4

Q: [Problem 2.12] A cycle in a graph is defined as a path originating and terminating at the same node. Show that there are no odd-length cycles in a d -dimensional hypercube.

A: A d -dimensional hypercube is a bipartite graph. Each node label has a parity determined by the sum of its bits. Every edge connects a node of even parity to a node of odd parity. Returning to the starting node requires an even number of bit flips, meaning all cycles must be of even length.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 5

Q: [Problem 2.13] The labels in a d -dimensional hypercube use d bits. Fixing any k of these bits, show that the nodes whose labels differ in the remaining $d - k$ bit positions form a $(d - k)$ -dimensional subcube composed of 2^{d-k} nodes.

A: Fixing k bits leaves $d - k$ bits free. Two nodes in this subset are adjacent if and only if their labels differ in exactly 1 of the remaining $d - k$ positions, which matches the definition of a $(d - k)$ -dimensional hypercube with 2^{d-k} nodes.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 6

Q: [Problem 2.14] Let A and B be two nodes in a d -dimensional hypercube. Let $H(A, B)$ be the Hamming distance between A and B , and $P(A, B)$ be the number of distinct parallel paths connecting A and B . Prove:

Minimum distance between A and B is $H(A, B)$.

Total number of parallel paths $P(A, B) = d$.

Number of parallel paths of length $H(A, B)$ is $H(A, B)$.

Length of remaining $d - H(A, B)$ parallel paths is $H(A, B) + 2$.

A:

Each link traversal flips 1 bit. Changing label A to B requires flipping $H(A, B)$ differing bits, so shortest path length = $H(A, B)$.

Node A has degree d (outgoing links in d dimensions), enabling d node-disjoint parallel paths.

Permuting the order of flipping the $H(A, B)$ differing bits produces $H(A, B)$ disjoint paths of minimum length $H(A, B)$.

For the $d - H(A, B)$ dimensions where A and B share identical bits, taking a step along any such dimension flips a matching bit (step away), requiring 1 step towards B and 1 step back, adding 2 extra hops to yield length $H(A, B) + 2$.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 7

Q: [Problem 2.15] Show that any partition of a d -dimensional hypercube into two equal halves of 2^{d-1} nodes will have at least 2^{d-1} direct links between them.

A: A d -dimensional hypercube is formed by connecting corresponding nodes of two $(d-1)$ -dimensional subcubes via 2^{d-1} cross links. By the edge-isoperimetric theorem on hypercubes, cutting 2^d nodes into two sets of 2^{d-1} nodes yields an edge cut of at least 2^{d-1} links.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 8

Q: [Problem 2.16] Determine the bisection width, diameter, and number of switching nodes and links for a reconfigurable mesh of p processing nodes. Compare it with a wraparound mesh.

A:

Diameter: $O(1)$ when dynamically reconfigured as a global bus, or $O(\sqrt{p})$ worst-case.

Bisection Width: $\sqrt{p}$.

Switching Nodes: p nodes (each having internal 15-pattern bus switches).

Communication Links: $2\sqrt{p}(\sqrt{p} - 1)$ grid links.

Pros/Cons: Allows $O(1)$ dynamic reconfigurable bus broadcasts, but only one node can broadcast per subbus at a time.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 9

Q: [Problem 2.17] Determine the bisection width, diameter, and total number of switching nodes in a $\sqrt{p} \times \sqrt{p}$ mesh of trees.

A:

Diameter: $4 \log_2(\sqrt{p}) = 2 \log_2 p$.

Bisection Width: $\sqrt{p}$.

Total Switching Nodes: $2\sqrt{p}(\sqrt{p} - 1) = 2p - 2\sqrt{p}$ intermediate tree nodes.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 10

Q: [Problem 2.20] Show how to fabricate a 2D wraparound mesh using wires of fixed length.

A: By interleaving the physical placement of nodes along rows and columns (interleaved/folded torus layout), nodes are arranged in permuted physical order (e.g., 0, 2, 1, 3), making all physical wire connections, including wraparound links, equal in length.

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 11

Q: [Problem 2.21] Show how to embed a p-node 3D mesh into a p-node hypercube. What are allowable values of p?

A: Use binary Reflected Gray Codes. A 3D mesh of size $2^r \times 2^s \times 2^t$ ($p = 2^{r+s+t}$) maps mesh node (i, j, k) to hypercube node $G(i, r-1) \parallel G(j, s-1) \parallel G(k, t-1)$. Neighboring mesh nodes map to hypercube nodes differing in 1 bit (dilation 1, congestion 1). Allowable values of p are powers of 2 ($p = 2^d$).

INTRODUCTION TO PARALLEL COMPUTING, CHAPTER 2 • QUESTION 12

Q: [Problem 2.25] Derive the diameter, number of links, and bisection width of a k-ary d-cube with $p = k^d$ nodes.

A:

Diameter: $d \cdot \lfloor k / 2 \rfloor$.

Number of Links: $d \cdot p = d \cdot k^d$.

Bisection Width: $2k^{d-1}$.